PMSCS PROGRAM

Department of Computer Science and Engineering

Jahangirnagar University

Date: 03/05/2024

Course Code: PMSCS-623P **Title:** Data Structures and Algorithms

Time: 3 Hours **TERM:** SPRING 2024 **Full Marks:** 60

<u>Answer any SIX from the following Questions</u>

1.

a) Choose the most *appropriate* answer: **[4]**

i) Given an array *arr* = {5, 6, 77, 88, 99} and key = 6; How many iterations are done until the element is found in Binary search Algorithm? a) 1 b) 3 c) 4 d) 2	**ii)** What is the best case for linear search? a) O(nlogn) b) O(logn) c) O(n) d) O(1)
iii) Given an array arr = {22, 27, 39, 50, 64, 69, 80} and key = 80; What are the mid values generated in the first and second iterations in Binary search Algorithm? a) 50 and 64 b) 50 and 69 c) 39 and 69 d) 39 and 69	**iv)** To perform Binary search, elements need NOT to be sorted. a) True b) False

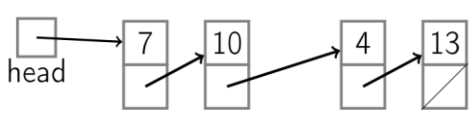
b) Apply **Buble Sort** algorithm to the following array: {-15, 11, 9, -2, 8, 4} to sort the array. **[4]**

c) Mention two important properties that an algorithm should always maintain. **[2]**

2.

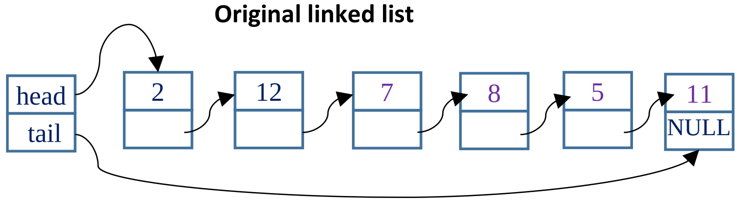
a) Define *linked list* with example. Mention any two key differences between *array* and *linked list*. **[1+2]**

b) Mention the time complexity for *PopFront* and *PushBack* operations for the following linked list. **[2]**



c) Consider the following <u>non-empty singly linked list</u> L of integers. We first like to remove the node containing the value 2 and then will remove the node containing the value of 12. Show the changes diagrammatically for each case separately. **[5]**

Original linked list

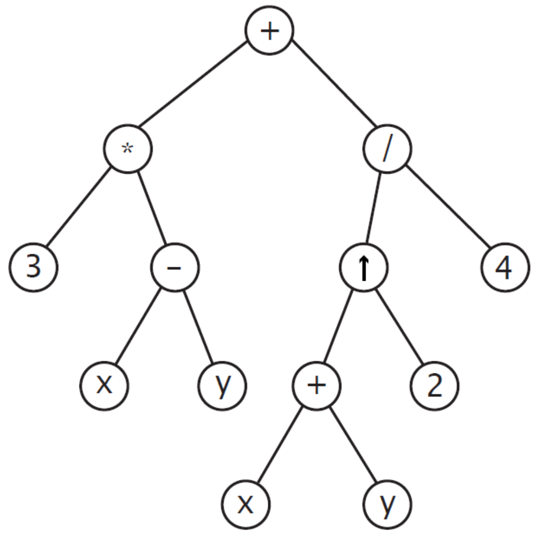


3.	a) Transform the following infix expression into its equivalent postfix expression using *stack*: ((a + b) – c) × (d / e) × f	**[4]**
	b) Find the value of the following arithmetic expression written in *postfix* notation using *stack*: 12, 7, 3, - , /, 2, 1, 5 , + , * , +	**[4]**
	c) What is *deque*? What is the difference between input-restricted and output-restricted deque?	**[1+1]**
4.	a) Construct the *table* and corresponding *labeled directed graph* using **Fast** *pattern matching* algorithm for the pattern ***P*** = *abababa*	**[2+4]**
	b) For each of the following patterns P and texts T, find the number **C** of *comparisons* to find the INDEX of *P* in *T* using the **Slow** algorithm. i) P=abc, T=$(ab)^5$ ii) P=abc, T=$(ab)^{2n}$ iv) P=aaa, T=$(aabb)^3$ iii) P=aaa, T=abaabbaaabbbaaaabbbb	**[4]**
5.	a) Perform traditional **Quicksort** algorithm to an array {3, 22, 15, 8, 19, 10, 13} to place 13 to its correct position.	**[3]**
	b) Derive the *worst case* time complexity for **Merge sort** algorithm.	**[3]**
	c) Briefly mention the procedure of how you can restore the heap property if a node violates the max heap property.	**[2]**

	d) Analyze the running time of the following algorithm and write it using ***O***() notation:
	`for(int i=1; i <=n; i++){`
	`    for(int j=(2 × n); j>=1; j=j--){`
	`        printf(“PMSCS 623P”);`
	`    }`
	`}`
	`for(k=n/4; k<=n/2; k=k++){`
	`            sum++;`
	`}`
		[2]

6.	a) Find the value of the *postfix* expression: 6 2 ↑ 9 / 7 2 3 ↑ + 5 / +	**[4]**
	b) Define Full Binary tree, Complete Binary tree and height of a rooted tree. Show example in each case.	**[3]**

c) Find the arithmetic expression (in **postfix** form) from the order rooted tree given below: **[3]**



7. a) A file has the following characters along with their frequencies: **[7]**

a:9, d:13, b:10, e:13, m:7, s:25, r: 23, q:16

Draw **Huffman** coding tree and compute the *optimal* Huffman code of each character mentioned above.

b) Given an infinite supply of each of the denominations {2, 5, 10, 20, 50, 100, 500, 1000} valued coins/notes, the task is to find the **minimum** number of coins and/or notes needed to make the change of amount **A = 226**? **[3]**

8. a) Compute the optimal price for the **0-1** *knapsack* problem for knapsack capacity, **W=8**. Which items will be in the optimal solution? What will be their total price? Show how you calculate the optimal solution using a matrix containing both price and arrows for each cell in the matrix. **[4]**

item	**A**	**B**	**C**	**D**	**E**
Weight	2	1	4	5	2
price	3	3	7	8	6

b) Compute the optimal solution for the **fractional** *knapsack* problem for knapsack capacity, **W=7**. Which items will be in the optimal solution with which weight each? What will be their total price? Which items do you choose in the 1st, 2nd, 3rd, … steps and why? **[6]**

item	**1**	**2**	**3**	**4**	**5**
Weight	3	1	2	3	2
price	17	7	10	15	19

N.B. Students must submit question along with answer scripts at the end of examination.